

5 Appendix: Proof of Proposition 1

Proof. Let $\mathbf{v} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be a velocity field. Define the explicit noise-to-image Euler update as

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \Delta t \mathbf{v}_{t_k}(\mathbf{x}_k), \quad t_{k+1} = t_k - \Delta t. \quad (11)$$

Define the velocity-reuse update for M consecutive steps using a previously evaluated velocity computed at $(t_{k_0}, \mathbf{x}_{k_0})$:

$$\tilde{\mathbf{x}}_{k+1} = \mathbf{x}_k - \Delta t \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0}), \quad k = k_0, \dots, k_0 + M - 1. \quad (12)$$

Assume Lipschitz continuity $L_x, L_t \geq 0$ such that for all relevant $\mathbf{x}, \mathbf{x}', t, t'$,

$$\|\mathbf{v}_t(\mathbf{x}) - \mathbf{v}_t(\mathbf{x}')\| \leq L_x \|\mathbf{x} - \mathbf{x}'\|, \quad \|\mathbf{v}_t(\mathbf{x}) - \mathbf{v}_{t'}(\mathbf{x})\| \leq L_t |t - t'|. \quad (13)$$

Single Step Error. Assume reuse the velocity at the first step $k_0 = k - 1$ gives error e_k ,

$$\begin{aligned} e_k &= \|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k-1}}(\mathbf{x}_{k-1})\| \\ &\leq \|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k-1}}(\mathbf{x}_k)\| + \|\mathbf{v}_{t_{k-1}}(\mathbf{x}_k) - \mathbf{v}_{t_{k-1}}(\mathbf{x}_{k-1})\| \\ &\leq L_t |t_k - t_{k-1}| + L_x \|\mathbf{x}_k - \mathbf{x}_{k-1}\| \\ &\leq L_t \Delta t + L_x \Delta t \|\mathbf{v}_{t_{k-1}}(\mathbf{x}_{k-1})\|. \end{aligned} \quad (14)$$

Then the error in $\tilde{\mathbf{x}}_{k+1}$ and $\mathbf{x}_{k+1}$ becomes

$$\begin{aligned} \|\mathbf{x}_{k+1} - \tilde{\mathbf{x}}_{k+1}\| &= \Delta t \|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k-1}}(\mathbf{x}_{k-1})\| \\ &\leq \Delta t^2 (L_t + L_x \|\mathbf{v}_{t_{k-1}}(\mathbf{x}_{k-1})\|). \end{aligned} \quad (15)$$

It shows that the reusing the velocity gives single step error of $O(\Delta t^2)$, which matches the error of standard Euler $x(t - \Delta t) = x(t) - \Delta t \dot{x} + O(\Delta t^2)$.

Accumulated Error. According to our proposed algorithm, we define standard single-step state update as

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \Delta t \mathbf{v}_{t_k}(\mathbf{x}_k) + \mathcal{DC}(\mathbf{x}_k - t_k \mathbf{v}_{t_k}(\mathbf{x}_k), \mathbf{y}, \mathbf{A}), \quad (16)$$

and the corresponding state update with velocity-reuse steps as

$$\tilde{\mathbf{x}}_{k+1} = \tilde{\mathbf{x}}_k - \Delta t \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0}) + \mathcal{DC}(\tilde{\mathbf{x}}_k - t_k \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0}), \mathbf{y}, \mathbf{A}), \quad k = k_0, \dots, k_0 + M - 1, \quad (17)$$

where $\mathcal{DC}(\mathbf{x}, \mathbf{y}, \mathbf{A})$ is the data consistency operation given measurement $\mathbf{y}$ and forward operator $\mathbf{A}$.

Here the same velocity $\mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0})$ is reused for M consecutive steps. Assume $\mathcal{DC}(\mathbf{x}, \mathbf{y}, \mathbf{A})$ is κ -non-expansive with respect to its image argument,

$$\|\mathcal{DC}(\mathbf{x}, \mathbf{y}, \mathbf{A}) - \mathcal{DC}(\mathbf{x}', \mathbf{y}, \mathbf{A})\| \leq \kappa \|\mathbf{x} - \mathbf{x}'\|, \quad 0 \leq \kappa \leq 1. \quad (18)$$

Let $\delta_k = \tilde{\mathbf{x}}_k - \mathbf{x}_k$. Then

$$\begin{aligned} \|\delta_{k+1}\| &\leq \|\delta_k\| + \Delta t \|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0})\| \\ &\quad + \kappa \|\delta_k - t_k (\mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0}) - \mathbf{v}_{t_k}(\mathbf{x}_k))\| \\ &\leq (1 + \kappa) \|\delta_k\| + (\Delta t + \kappa t_k) \underbrace{\|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0})\|}_{\text{velocity mismatch}}. \end{aligned} \quad (19)$$

Using Lipschitz continuity, we have

$$\underbrace{\|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0})\|}_{\text{velocity mismatch}} \leq L_x \underbrace{\|\mathbf{x}_k - \mathbf{x}_{k_0}\|}_{\text{state drift}} + L_t |t_k - t_{k_0}|. \quad (20)$$

Assume the practical updates have bounded increments,

$$\|\mathbf{x}_{j+1} - \mathbf{x}_j\| \leq \Delta t U_{\max}, \quad (21)$$

which holds when both the velocity field and the DC correction remain bounded along the trajectory. Then

$$\begin{aligned} \underbrace{\|\mathbf{x}_k - \mathbf{x}_{k_0}\|}_{\text{state drift}} &\leq \sum_{j=k_0}^{k-1} \|\mathbf{x}_{j+1} - \mathbf{x}_j\| \\ &\leq (k - k_0) \Delta t U_{\max}. \end{aligned} \quad (22)$$

Since $|t_k - t_{k_0}| = (k - k_0) \Delta t$, we obtain

$$\underbrace{\|\mathbf{v}_{t_k}(\mathbf{x}_k) - \mathbf{v}_{t_{k_0}}(\mathbf{x}_{k_0})\|}_{\text{velocity mismatch}} \leq (k - k_0) \Delta t (L_t + L_x U_{\max}). \quad (23)$$

Let $C = L_t + L_x V_{\max}$ and $t_{\max} = \max_k t_k$. Then

$$\|\delta_{k+1}\| \leq (1 + \kappa) \|\delta_k\| + (\Delta t + \kappa t_{\max})(k - k_0) \Delta t C. \quad (24)$$

Unrolling over M consecutive reuse steps with $\delta_{k_0} = 0$ gives

$$\begin{aligned} \underbrace{\|\delta_{k_0+M}\|}_{\text{accumulated deviation}} &\leq (\Delta t + \kappa t_{\max}) C \Delta t \sum_{j=0}^{M-1} (1 + \kappa)^{M-1-j} j \\ &\leq (\Delta t + \kappa t_{\max}) C \Delta t (1 + \kappa)^{M-1} \frac{M(M-1)}{2}. \end{aligned} \quad (25)$$

Therefore for bounded M ,

$$\|\delta_{k_0+M}\| = O(M^2 \Delta t), \quad (26)$$

i.e. the accumulated error remains controlled for bounded consecutive reuse steps.